1. Evaluating All Options At every step (starting with the very first root node), the algorithm looks at all your available features (e.g., age, income, weather conditions) and evaluates every possible way to split the data based on those features.

2. Scoring the Splits (The Math) It calculates a mathematical score for each possible split to see which one creates the most organized or "purest" resulting groups.

- **For Classification (predicting categories like Yes/No):** It typically uses metrics like **Gini Impurity** or **Entropy (Information Gain)** to measure how well a split separates the different classes.
- **For Regression (predicting continuous numbers like price):** It uses metrics like **Mean Squared Error (MSE)** to see which split reduces the variance the most.

3. Making the Branch The algorithm automatically selects the single feature and specific threshold (e.g., "Age > 30" or "Income < $50,000") that yields the absolute best mathematical score. It then creates a branch based on that winning rule.

4. Repeating the Process It moves down to the newly created sub-nodes and repeats the exact same evaluation and scoring process to create the next set of branches.

When does it stop branching? By default, the algorithm will keep branching automatically until every final leaf node is 100% pure (meaning all the data points in that specific branch belong to the exact same category).

However, letting a tree grow until it is 100% pure usually leads to **overfitting** (memorizing the training data instead of learning general patterns). To control the automatic branching, data scientists manually set restrictions known as **hyperparameters** before training begins. Common restrictions include:

- **Max Depth:** "Stop branching automatically once the tree reaches 5 levels deep."
- **Min Samples Split:** "Do not create a new branch if a node has fewer than 20 data points left in it."

In short, the algorithm automatically discovers the best rules and creates the branches, but you control the boundaries of how large the tree is allowed to grow.